

DATE	30 September 2026
TEAM ID	SWUID2026349965
PROJECT NAME	PocketSmart AI: Your Smart Budget & Recommendation Assistant

Phase 03: Project Design

1. System Architecture

PocketSmart AI follows a layered, modular architecture. The browser communicates with a FastAPI backend that handles authentication, prompt preparation, Gemini 2.5 calls and shopping link generation.

Layer	Component	Responsibility
Presentation	Jinja2 templates and CSS/JavaScript	Collect inputs and display results as cards and tables
API	FastAPI application with CORS middleware	Routing, validation, token check, static files
Service	Recommendation utility functions	Prompt preparation, Gemini calls, JSON parsing, link building
Model	Pydantic schemas	Input and output data structures
AI	Google Gemini 2.5 (text and image)	Budget allocation and product suggestions
Storage	Upload folder, in-memory users, sessions and history	Images and per-user data

1.1 Request Flow

- The user submits a planner form from the browser.
- FastAPI checks the access token and identifies the current user.
- The last budget details are saved in the user's session data.
- A structured prompt is prepared; for jewelry, the outfit image is attached.
- Gemini 2.5 returns a JSON plan with categories, items, prices and search terms.
- The backend adds calculation tables and shopping links per item.
- The result is saved to history and returned to the browser for display.

1.2 Cloud Reference Architecture

The project brief describes an AWS-based deployment: users reach the application through CloudFront and Route 53; API Gateway exposes REST endpoints that invoke a Lambda function; the Lambda sends prompts to Gemini and fetches product details from e-commerce platforms; Secrets Manager holds API keys; IAM controls access; S3 stores uploaded images and static assets; DynamoDB stores users, preferences, budgets and history; CloudWatch handles logs and metrics.

2. Route Design

Method	Route	Description	Login needed
GET	/	Landing page	No
GET	/login and /register	Login and registration pages	No
POST	/token	Verify credentials and issue access token	No
POST	/logout	Invalidate token and end session	Yes
GET	/dashboard	Dashboard with planner cards and recent activity	Yes
GET	/home-planner, /party-planner, /jewelry-planner	Planner form pages	Yes
POST	/home-budget	Home recommendations	Yes
POST	/party-budget	Party recommendations	Yes
POST	/jewelry-budget	Jewelry recommendations with optional image	Yes
GET	/recommendation-history	List of saved plans, newest first	Yes
GET	/recommendation-details/{id}	Full details of one saved plan	Yes
GET	/history	History page	Yes
GET, POST	/session-info, /session-data	Read or update session information	Yes

The project brief refers to the planner routes as generate-home, generate-party and generate-jewelry. In the implemented application these are available as home-budget, party-budget and jewelry-budget; both names describe the same functionality.

3. Data and Prompt Design

3.1 Structure of AI Responses

Planner	Main parts of the response
Home	Total budget; budget breakdown by category with allocation and items (name, description, estimated price, quantity, search terms); calculation table; remaining budget; additional suggestions
Party	Same breakdown structure plus venue suggestions (name, type, capacity, estimated cost, search terms); calculation table in INR built by the backend
Jewelry	Optional outfit analysis (colours, style, formality); jewelry items (type, description, style, estimated price, search terms); remaining budget; styling tips

3.2 Prompt Design Principles

Principle	How it is applied
State the market	The prompt states that the user is in India and that prices must be in INR.
List every input	Budget, counts, rooms, party details, occasion and notes are all included.
Force a structure	The model is told to answer in a fixed JSON structure.
Add hard constraints	The model must keep the total within the budget.
Ask for search terms	Each item includes a search phrase for Indian shopping platforms.
Use the image	For jewelry, the outfit image is sent with the text so colour and style are considered.

3.3 Category to Platform Mapping (Party Planner)

Category	Platforms linked
Venue	Google, Booking.com, MakeMyTrip, OYO, NoBroker
Catering and food	Swiggy, Zomato, BigBasket
Decoration	Amazon, Flipkart, Meesho, Myntra
Entertainment	BookMyShow, Amazon, Flipkart
Other categories	Amazon, Flipkart, Google

4. User Interface and Security Design

4.1 Screen Inventory

Screen	Main elements
Landing page	Hero section, planner cards, testimonials, footer
Register and Login	Username, email, password fields; links between the pages
Dashboard	Welcome banner, three planner cards, recent activity, history button
Home planner	Budget, counts, room selection, notes, generate button
Party planner	Budget, guests, party type, venue type, needs, notes
Jewelry planner	Budget, occasion, style text, outfit image upload with preview
Result pages	Budget summary, category cards with item tables, shopping link buttons, suggestions
History	Cards for each plan with totals, remaining budget and full-details button

4.2 Design Guidelines

- Clean card-based layout with an icon for each planner.
- Consistent navigation bar: Dashboard, planners, History, Logout.
- Responsive grid so cards stack neatly on small screens.
- Clear labels, sensible defaults and helpful placeholder text.

4.3 Security Design

Concern	Decision
Passwords	Hashed with bcrypt
Access tokens	Signed tokens with 30-minute expiry, stored in an HTTP-only cookie
Logout	Token is blacklisted and the session is deleted
Session cleanup	A background task removes sessions idle for more than 30 minutes
Secrets	API key and secret key are read from an environment file
CORS	Open during development; restricted to trusted origins in production